

10-Class Fundus Eye Disease Classification: Research & Progress Report

Department of Computer Science & Engineering, Jadavpur University
B.Tech Major Project (8th Semester)

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1. Dataset Setup & Experimental Protocol

Following supervisor directives, the dataset has been structured into a **pure folder-based hierarchy (zero CSV files)** with an **exactly balanced distribution** of 400 images per class across all 10 ophthalmic disease classes (4,000 total images):

Split	Images per Class	Total Images (10 Classes)	Protocol Purpose & Methodology
Training Set	140	1,400	Dynamic Albumentations data augmentation (rotations, affine, color jitter, cropping).
Validation Set	130	1,300	Model checkpoint selection based on Macro F1 to prevent overfitting.
Test Set	130	1,300	Independent untouched test evaluation ensuring high statistical confidence.
Total Dataset	400 per class	4,000 Total	100% Class Balanced across all 10 diagnostic categories.

2. SOTA Literature Matrix & Our Models' Performance

Model / Study	Architecture / Type	Test Accuracy	Macro F1	ROC-AUC	Cohen's Kappa	Key Notes
Rashid et al. (2021)	ResNet-50 / DenseNet-121	83.40%	82.80%	0.9650	~0.90	Initial Mendeley baseline benchmark.
PLoS ONE (2023)	Swin Transformer (Swin-T)	86.80%	86.20%	0.9780	~0.91	Shifted window self-attention.
EyeFusionNet (2023)	DenseNet-169 + TNT	89.20%	88.70%	0.9840	~0.93	Dual-branch CNN + Transformer fusion.
RETFound (Nature '23)	ViT-Large/16 (1.6M Pretrain)	88.5 - 91.2%	89.40%	0.9880	~0.94	Moorfields/UCL foundation model.
Our Model (EXP-001) ■ CNN Baseline	EfficientNet-B0 + Focal Loss	83.38%	83.11%	0.9774	0.9115	Evaluated on 1,300 test images. 98.15% specificity.
Our Model (EXP-002) ■ Foundation Model	Microsoft BiomedCLIP (ViT-B/16)	83.85%	83.69%	0.9802	0.9128	Fine-tuned on 15M PubMed weights; beats CNN baseline on all metrics.

3. Detailed Per-Disease Test Performance (1,300 Images Evaluated)

Below is the exact class-by-class sensitivity, specificity, and F1-score comparison for both evaluated architectures on the 1,300 test images (130 samples per class):

Disease Category (10 Classes)	Test Samples	EffNet Sensitivity	EffNet Specificity	EffNet F1	BiomedCL IP Sensitivity	BiomedCL IP Specificity	BiomedCL IP F1	Winner / Gain
Pterygium	130	100.0%	100.0%	1.0000	100.0%	100.0%	1.0000	Tie (Perfect)
Retinal Detachment	130	96.92%	99.66%	0.9692	100.0%	99.83%	0.9924	BiomedCLIP (+2.3%)
Retinitis Pigmentosa	130	97.69%	98.80%	0.9373	96.15%	99.57%	0.9615	BiomedCLIP (+2.4%)
Disc Edema	130	90.00%	99.57%	0.9286	95.38%	99.06%	0.9358	BiomedCLIP (+0.7%)

Diabetic Retinopathy	130	86.92%	98.46%	0.8659	86.15%	99.15%	0.8889	BiomedCLIP (+2.3%)
CSCR [Color Fundus]	130	93.85%	98.29%	0.8971	80.77%	98.46%	0.8300	EffNet-B0
Myopia	130	70.00%	97.95%	0.7429	80.00%	97.44%	0.7879	BiomedCLIP (+4.5%)
Healthy	130	79.23%	95.81%	0.7305	83.08%	95.56%	0.7448	BiomedCLIP (+1.4%)
Macular Scar	130	71.54%	97.01%	0.7209	66.15%	96.75%	0.6772	EffNet-B0
Glaucoma	130	47.69%	95.98%	0.5188	50.77%	96.24%	0.5500	BiomedCLIP (+3.1%)
Macro Average / Overall	1,300	83.38%	98.15%	0.8311	83.85%	98.21%	0.8369	BiomedCLIP Wins

4. Key Scientific Insights for Supervisor Review

- 1. **Superiority of Medical Pretraining:** Fine-tuning Microsoft BiomedCLIP (pretrained on 15M PubMed biomedical image-text pairs) boosted overall test accuracy to **83.85%**, Macro F1 to **83.69%**, and ROC-AUC to **0.9802**, outperforming standard ImageNet CNN transfer learning.
- 2. **High Specificity (98.21%):** Both models exhibit very low false-positive rates, crucial for real-world ophthalmic screening pipelines.
- 3. **Diagnosing Glaucoma remains the Core Bottleneck:** Glaucoma sensitivity is the lowest (~50.8%), caused by subtle cup-to-disc ratio changes. This establishes the clear rationale for our next architectural novelty.

5. Proposed Next Stage: Novel Attention Enhancement (Targeting 88%+)

To exceed standalone baselines and push accuracy toward 90%, we propose integrating a **Convolutional Block Attention Module (CBAM)** and **Multi-Scale Feature Pyramid Fusion** into the BiomedCLIP visual backbone, providing focused attention on the optic cup/disc and macula.